

Durham College

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INFT 1103: Operating System Security

Lab Week 8: Examine a Linux Server

November 11, 2025

Part 1: Basic Linux Administration

1. Download Ubuntu Server 22.04 from here: <https://ubuntu.com/download/server> (choose “manual server installation” to get the download file)
2. Create a new VM using the Ubuntu Server 22.04 iso
 - a. For the username setup, use your name
3. Once the server is installed, login with your created username (your name)
4. Once you are logged in, use the command “cd /”
5. Use the command “ls -la”
 - a. Take a screenshot and paste it here:

```
melvin@ubuntu:~$ cd /
melvin@ubuntu:/$ ls -la
total 76
drwxr-xr-x 20 root root 4096 Nov  3 18:54 .
drwxr-xr-x 20 root root 4096 Nov  3 18:54 ..
lrwxrwxrwx  1 root root    7 Sep 11 2024 bin -> usr/bin
drwxr-xr-x  4 root root 4096 Nov  3 18:58 boot
dr-xr-xr-x  2 root root 4096 Sep 11 2024 cdrom
drwxr-xr-x 20 root root 4100 Nov  3 18:58 dev
drwxr-xr-x 96 root root 4096 Nov  3 18:58 etc
drwxr-xr-x  3 root root 4096 Nov  3 18:58 home
lrwxrwxrwx  1 root root    7 Sep 11 2024 lib -> usr/lib
lrwxrwxrwx  1 root root    9 Sep 11 2024 lib32 -> usr/lib32
lrwxrwxrwx  1 root root    9 Sep 11 2024 lib64 -> usr/lib64
lrwxrwxrwx  1 root root   10 Sep 11 2024 libx32 -> usr/libx32
drwx----- 2 root root 16384 Nov  3 18:55 lost+found
drwxr-xr-x  2 root root 4096 Sep 11 2024 media
drwxr-xr-x  2 root root 4096 Sep 11 2024 mnt
drwxr-xr-x  2 root root 4096 Sep 11 2024 opt
dr-xr-xr-x 300 root root    0 Nov  3 18:58 proc
drwx----- 4 root root 4096 Nov  3 18:58 root
drwxr-xr-x 29 root root  820 Nov  3 18:59 run
lrwxrwxrwx  1 root root    8 Sep 11 2024/sbin -> usr/sbin
drwxr-xr-x  6 root root 4096 Sep 11 2024 snap
drwxr-xr-x  2 root root 4096 Sep 11 2024 srv
dr-xr-xr-x 13 root root    0 Nov  3 18:58 sys
drwxrwxrwt 14 root root 4096 Nov  3 18:59 tmp
drwxr-xr-x 14 root root 4096 Sep 11 2024 usr
drwxr-xr-x 13 root root 4096 Sep 11 2024 var
melvin@ubuntu:/$ _
```

6. Use the command "locale"

a. Take a screenshot and paste it here:

```
melvin@ubuntu:/$ locale
LANG=en_US.UTF-8
LANGUAGE=
LC_CTYPE="en_US.UTF-8"
LC_NUMERIC="en_US.UTF-8"
LC_TIME="en_US.UTF-8"
LC_COLLATE="en_US.UTF-8"
LC_MONETARY="en_US.UTF-8"
LC_MESSAGES="en_US.UTF-8"
LC_PAPER="en_US.UTF-8"
LC_NAME="en_US.UTF-8"
LC_ADDRESS="en_US.UTF-8"
LC_TELEPHONE="en_US.UTF-8"
LC_MEASUREMENT="en_US.UTF-8"
LC_IDENTIFICATION="en_US.UTF-8"
LC_ALL=
melvin@ubuntu:/$ _
```

7. Create a local user account. Use the command (without the brackets): *sudo adduser [your favorite movie character]*

8. Use the command *cd /home*

a. Take a screenshot and paste it here:

```
melvin@ubuntu:/$ sudo adduser mumble
[sudo] password for melvin:
Adding user `mumble' ...
Adding new group `mumble' (1001) ...
Adding new user `mumble' (1001) with group `mumble' ...
Creating home directory `/home/mumble' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for mumble
Enter the new value, or press ENTER for the default
  Full Name []:
  Room Number []:
  Work Phone []:
  Home Phone []:
  Other []:
Is the information correct? [Y/n] Y
melvin@ubuntu:/$ cd /home
melvin@ubuntu:/home$ _
```

9. Use the command `sudo chmod 777 /home/[movie character]`
 - a. Use the command `ls -ld /home/`
 - b. What does the `chmod 777` command do?

The `chmod 777` command grants the most permissions to the specified user or users for a file directory. The digits in 777 represent the permissions for three categories. The first digit grants permissions to the file's owner, the second digit to the group associated with the file, and the third digit to other users on the system. The number 7 is the sum of the permission values 4 (read), 2 (write), and 1 (execute).

- c. Take screenshot and paste it here:

```
melvin@ubuntu:/home$ sudo chmod 777 /home/mumble
melvin@ubuntu:/home$ ls -ld /home/
drwxr-xr-x 4 root root 4096 Nov  3 19:03 /home/
melvin@ubuntu:/home$ _
```

10. Use the command `cd /home/[movie character]`
11. Create a directory using the command `mkdir INFT1103files`
12. Browse to that directory
13. Create a file within the INFT1103 directory using the command `touch INFT1103file`
14. Use the command `ls -la` to confirm the file permissions
 - a. Take screenshot and paste it here:

```
melvin@ubuntu:/home/mumble$ cd /home/mumble/INFT1103files
melvin@ubuntu:/home/mumble/INFT1103files$ touch INFT1103file
melvin@ubuntu:/home/mumble/INFT1103files$ ls -la
total 8
drwxrwxr-x 2 melvin melvin 4096 Nov  3 19:13 .
drwxrwxrwx 3 mumble mumble 4096 Nov  3 19:12 ..
-rw-rw-r-- 1 melvin melvin    0 Nov  3 19:13 INFT1103file
melvin@ubuntu:/home/mumble/INFT1103files$
```

15. Change the file permissions on the `INFT1103file` as follows:
 - a. Owner: rwx
 - b. Group: x
 - c. Other: none
 - i. What command did you use?

`chmod 710 INFT1103file`

16. Use the command `ls -la` to show the files contained in your directory and their permissions.

a. Take a screenshot and paste it here:

```
melvin@ubuntu:/home/mumble/INFT1103files$ chmod 710 INFT1103file
melvin@ubuntu:/home/mumble/INFT1103files$ ls -la
total 8
drwxrwxr-x 2 melvin melvin 4096 Nov  3 19:13 .
drwxrwxrwx 3 mumble mumble 4096 Nov  3 19:12 ..
-rwx--x--- 1 melvin melvin    0 Nov  3 19:13 INFT1103file
melvin@ubuntu:/home/mumble/INFT1103files$
```